AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Amity University Uttar Pradesh, Greater Noida

Minor Project (ETMN100) — WEEKLY PROGRESS REPORT (WPR) – 7

Week: 31/08/2026 – 06/09/2026	**Enrollment No:** A41105223039
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Faculty Guide: Dr. Girish Paliwal	**Topic:** Discussion & Comparative Analysis
Project: WaterTriage: Data-Driven Water Quality Risk Scoring for UP & Bihar	**Status:** On Schedule

TARGETS SET FOR THE WEEK

- Perform comparative hydrochemical synthesis between Uttar Pradesh and Bihar across 1,359 lab samples and 5,718 parameter readings.
- Mathematically evaluate and validate the Acute Hazard Escalation Rule against conventional arithmetic WQI dilution.
- Analyze temporal recurrence metrics across 100,690 historical contamination events (2009–2012) to identify persistent aquifer failure epicenters.
- Synthesize actionable engineering and policy intervention matrix for Jal Jeevan Mission (JJM), detailing CapEx, OpEx, and feasibility.

ACHIEVEMENTS FOR THE WEEK

1. Comparative Hydrochemical Analysis: The Indo-Gangetic Divide

Analysis of 1,359 validated samples (747 UP, 612 Bihar) containing 5,718 discrete readings confirms distinct hydrochemical regimes across the Gangetic basin. UP exhibits severe geogenic Fluoride ($F^- > 1.5$ mg/L in Unnao/Firozabad) and non-point agricultural Nitrate ($ ext{NO}_3^- > 45$ mg/L). In contrast, Bihar's reducing alluvial aquifers exhibit pervasive dissolved Iron ($ ext{Fe} > 2.5$ mg/L in Purnia/Katihar) and localized floodplain Arsenic ($ ext{As} > 0.05$ mg/L along Buxar-Bhagalpur belt).

Metric / Parameter	Uttar Pradesh (UP)	Bihar	Regional Hydrochemical Determinants
Total Scored Samples	747	612	Live WQMIS ingestion across priority districts
Mean Composite Risk (CRS)	14.30	8.83	UP has broader co-occurrence of multiple contaminants
Critical Band (CRS ≥ 75)	14 (1.9%)	8 (1.3%)	Acute pathogens (E. coli) & toxic Arsenic triggers
High Severity (50 ≤ CRS < 75)	337 (45.1%)	156 (25.5%)	Geogenic Fluoride, Iron, & agricultural Nitrates
Low / Compliant (CRS < 25)	385 (51.5%)	448 (73.2%)	Baseline compliant drinking water sources
Dominant Contaminant	Fluoride (F■) & Nitrate (NO■■)	Iron (Fe) & Arsenic (As)	Basement granites vs Holocene reducing floodplains

2. Validation of the Acute Hazard Escalation Invariant

Conventional arithmetic Water Quality Indices suffer from the *dilution (eclipsing) flaw*: a single lethal toxin (e.g., Arsenic at 0.08 mg/L, 8× BIS limit) is mathematically masked if other parameters (pH, TDS, Turbidity) are normal, yielding a misleading 'Good' score (~32/100). WaterTriage enforces an **Acute Hazard Escalation Invariant**: any Category-1 carcinogen (Arsenic ≥ 0.05 mg/L) or acute pathogen (E. coli > 0) immediately forces the composite score to Critical (CRS ≥ 75.0). This invariant rescued **22 habitations** across UP and Bihar from dangerous false-negative classification.

3. Recurrence Dynamics & Chronic Hotspots: Khokha Case Study

Evaluating 100,690 historical contamination records (2009–2012) across 14,044 villages demonstrates that snapshot sampling fails to capture structural aquifer failure. In **Khokha village (Purnia, Bihar)**, 166 recurring Iron exceedances were recorded over multi-year cycles. While Iron is normally low-weighted (aesthetic nuisance), 166 repeated breaches confirm irreversible shallow aquifer lateritization. Local officials repeatedly drilled shallow handpumps that failed; the system output mandates deep confined aquifer tapping (>180m) or piped surface water.

4. Jal Jeevan Mission (JJM) Policy & Engineering Intervention Matrix

WaterTriage bridges data science and civil engineering by translating risk scores directly into actionable infrastructure recommendations categorized by CapEx, OpEx, and institutional dispatch protocols:

Contaminant Profile	Target Region	Engineering Remedy	CapEx vs OpEx	Institutional Action
Acute Pathogen (E. coli > 0)	Shallow handpumps (Lucknow, Hardoi)	Inline electro-chlorination & sanitary wellhead grouting	Low CapEx (■15k/pump) Low OpEx	Immediate PHED disinfection squad; village boiling advisory
Geogenic Fluoride (F■ > 1.5 mg/L)	Central UP (Unnao, Firozabad)	Activated Alumina Units / CSIR-NEERI Nalgonda plants	Med CapEx (■2.5–5L) Med OpEx (alum refill)	Deploy community defluoridation; VWSC water testing kits
Dissolved Iron (Fe > 1.0 mg/L)	North Bihar (Purnia, Katihar)	Aeration-Sedimentation-Filtration (IRP) gravity sand units	Low-Med CapEx (■1–2L) Very Low OpEx	Attach top-head sand filters to existing handpumps
Geogenic Arsenic (As ≥ 0.05 mg/L)	Ganga floodplains (Buxar, Bhagalpur)	Adsorptive Media Filters (CSIR-AMPRI) & Deep Aquifer Bores	High CapEx (■8–15L) High OpEx (toxic media)	Immediate shallow well seal; shift to Multi-Village Piped Scheme
Nitrate Excess (NO■■ > 45 mg/L)	Agrarian Western & Central UP	Watershed agricultural buffer zones & RO drinking kiosks	High CapEx (■5–8L) High OpEx (reject brine)	Fertilizer capping policy; centralized community RO kiosk

5. Literature Alignment & Limitations (Reflecting on WPR-4)

• **BIS IS 10500:2012 Compliance:** Scoring functions strictly mirror Indian statutory limits distinguishing acceptable vs permissible bounds.
• **Hydrogeological Grounding:** Co-occurrence of Arsenic and Iron in Holocene sediments aligns with Chakraborti et al. (2018) and CGWB Gangetic Basin surveys.
• **Acknowledged Scope Limitations:** The 1,359 samples represent prioritized geographic clusters rather than uniform coverage. Furthermore, while the live dashboard (port 3000) visualizes 121 districts, automated scraping remains rate-limited to avoid burdening government servers.

FUTURE WORK PLANS

- **WPR-8 Deliverable (Conclusion & Limitations):** Finalize system limitations, technical trade-offs, and multi-year maintenance framework for the final project report.
- **Automated Work-Order Generator:** Build one-click PDF work-order rosters for Gram Panchayat and block junior engineers directly from priority queue queries.
- **Predictive Spatial Kriging:** Prototype spatial interpolation to estimate contamination probabilities for unmonitored habitations between surveyed clusters.

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